

Student Learning & AI Usage Impact Analysis

1. Project Overview

This project analyzes the progress in students learning by AI usage. The dataset includes details of 50000 distinct students from different domains. The goal is to establish a relationship between AI usage and different factors such as post-semester CGPA, retention score, etc.

2. Dataset Summary

The dataset is imported from Kaggle.

- Rows: 50000
- Columns: 16
- Key features:
 - Student Demographics (Student ID , Major Category , Year of Study)
 - Academic Performance (Pre-Semester CGPA , Post-Semester CGPA , Skill Retention Score)
 - AI Usage & Learning Behavior (Weekly Gen AI Hours, Primary AI Use Case , Number of AI Tools Used , Paid Subscription , Prompt Engineering Skill)
 - Student Well-Being (Anxiety Level , Burnout Risk , Perceived AI Dependency)

3. Exploratory Data Analysis using Python

Began with data preparation and cleaning in Python.

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used `ai.info()` to check structure and `ai.describe()` for summary statistics.

```
ai.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 16 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Student_ID                           50000 non-null  int64
 1   Major_Category                       50000 non-null  object
 2   Year_of_Study                        50000 non-null  object
 3   Pre_Semester_GPA                     50000 non-null  float64
 4   Weekly_GenAI_Hours                   50000 non-null  float64
 5   Primary_Use_Case                     50000 non-null  object
 6   Prompt_Engineering_Skill             50000 non-null  object
 7   Tool_Diversity                       50000 non-null  int64
 8   Paid_Subscription                    50000 non-null  bool
 9   Traditional_Study_Hours              50000 non-null  float64
10   Perceived_AI_Dependency              50000 non-null  int64
11   Institutional_Policy                 50000 non-null  object
12   Anxiety_Level_During_Exams          50000 non-null  int64
13   Post_Semester_GPA                   50000 non-null  float64
14   Skill_Retention_Score                50000 non-null  float64
15   Burnout_Risk_Level                   50000 non-null  object
dtypes: bool(1), float64(5), int64(4), object(6)
memory usage: 5.8+ MB
```

ai.describe(include= 'all')									
	Student_ID	Major_Category	Year_of_Study	Pre_Semester_GPA	Weekly_GenAI_Hours	Primary_Use_Case	Prompt_Engineering_Skill	Tool_Diversity	Paid_Subscription
count	50000.000000	50000	50000	50000.000000	50000.000000	50000	50000	50000.000000	50000
unique	NaN	5	5	NaN	NaN	5	3	NaN	2
top	NaN	STEM	Junior	NaN	NaN	Debugging/Troubleshooting	Beginner	NaN	False
freq	NaN	15059	11045	NaN	NaN	12295	18495	NaN	28846
mean	125000.500000	NaN	NaN	3.146102	8.427752	NaN	NaN	2.80026	NaN
std	14433.901067	NaN	NaN	0.478854	8.269490	NaN	NaN	1.18802	NaN
min	100001.000000	NaN	NaN	1.183000	0.000000	NaN	NaN	1.00000	NaN
25%	112500.750000	NaN	NaN	2.834000	2.390000	NaN	NaN	2.00000	NaN
50%	125000.500000	NaN	NaN	3.210000	5.800000	NaN	NaN	3.00000	NaN
75%	137500.250000	NaN	NaN	3.521000	11.720000	NaN	NaN	4.00000	NaN
max	150000.000000	NaN	NaN	3.998000	40.000000	NaN	NaN	5.00000	NaN

Traditional_Study_Hours	Perceived_AI_Dependency	Institutional_Policy	Anxiety_Level_During_Exams	Post_Semester_GPA	Skill_Retention_Score	Burnout_Risk_Level
50000.000000	50000.000000	50000	50000.000000	50000.000000	50000.000000	50000
NaN	NaN	3	NaN	NaN	NaN	3
NaN	NaN	Allowed_With_Citation	NaN	NaN	NaN	Medium
NaN	NaN	25224	NaN	NaN	NaN	21144
11.209271	3.505360	NaN	4.270760	3.349299	75.798125	NaN
5.156426	1.820812	NaN	2.144066	0.495673	13.281626	NaN
1.000000	1.000000	NaN	1.000000	1.000000	10.780000	NaN
7.560000	2.000000	NaN	3.000000	3.023750	66.820000	NaN
11.180000	3.000000	NaN	4.000000	3.421000	76.000000	NaN
14.710000	5.000000	NaN	6.000000	3.749000	85.190000	NaN
35.860000	10.000000	NaN	10.000000	4.000000	100.000000	NaN

- **Missing Data:** Checked for missing data by using ai.isnull().sum(). There were no missing values.
- **Column Standardization:** Renamed columns into lowercase for better documentation.
- **Feature Engineering:**
 - Created performance_status column by using post and pre semester cgpa.
 - Created anxiety_level column by categorizing the score into low , medium and high.
- **Database Integration :** Connected Python script to PostgreSQL and loaded the cleaned Dataframe into the database for SQL analysis.

4. Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key questions.

1. **Number of Students by Major Category** – Found number of students in particular major category.

	major_category text	number_of_students bigint
1	STEM	15059
2	Business	12538
3	Humanities	9994
4	Medical	6476
5	Arts	5933

2. **Average Pre and Post Semester Cgpa By Major Category** – Figured out average of pre and post semester cgpa for different categories.

	major_category text	average pre-semester gpa numeric	average post-semester gpa numeric
1	Business	3.14	3.34
2	STEM	3.15	3.36
3	Medical	3.15	3.35
4	Arts	3.15	3.34
5	Humanities	3.15	3.35

3. **Impact of AI usage** – Identified number of students who improved their performance.

	performance_status text	number_of_students bigint
1	Improved	43759
2	Declined	6192
3	No Change	49

4. **Average Weekly Hours spend by each category on GenAI** –

	major_category text	average_weekly_genAI_hours numeric
1	STEM	10.49
2	Business	8.28
3	Medical	7.55
4	Arts	7.27
5	Humanities	6.77

5. **Average Time Spent by Paid and Non-Paid Users**- Identified the number of students who spend more average time on GenAI from the group(paid & non-paid users).

	paid_subscription boolean	average_weekly_genai_hours numeric
1	true	10.33
2	false	7.03

6. **Students facing High Anxiety Level by Major Category-** Identified total number of students from each major category who are facing high anxiety level.

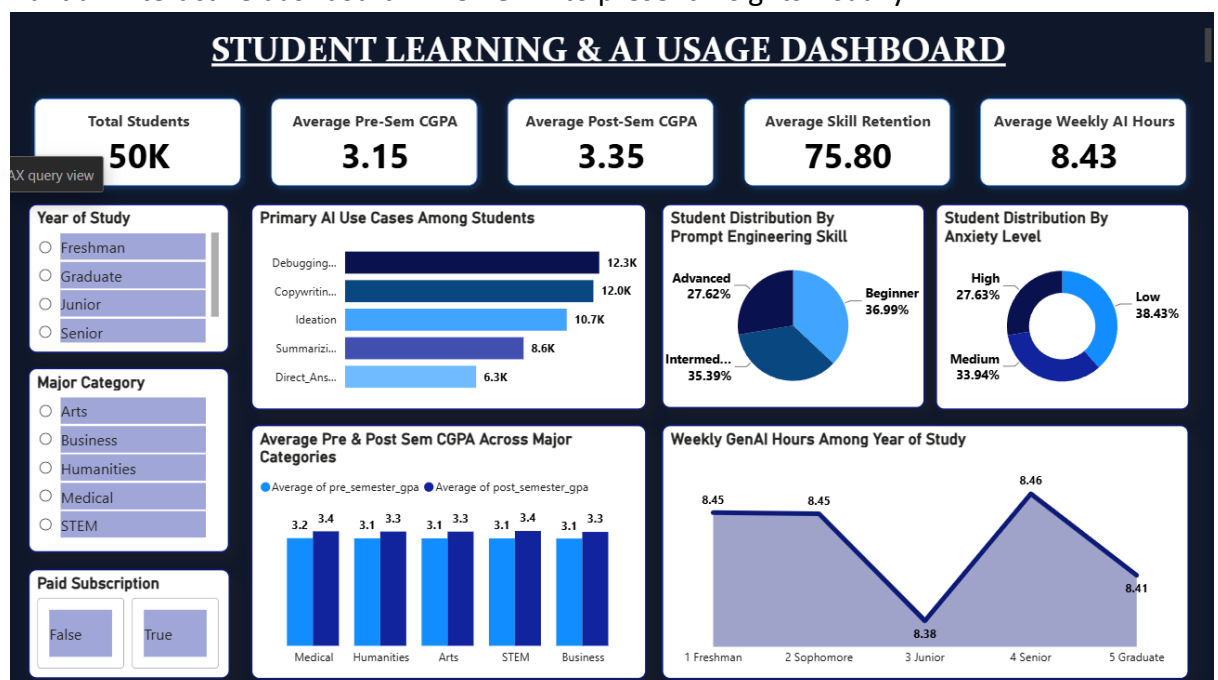
	major_category text	number_of_student bigint
1	STEM	9573
2	Business	7686
3	Humanities	5973
4	Medical	3957
5	Arts	3598

7. **Major category contributing the highest number of students to the overall top 20 students**

	major_category text
1	STEM

5. Dashboard in Power BI

Built an interactive dashboard in **Power BI** to present insights visually.



6. Insights

- The average post semester CGPA was higher than the pre semester CGPA, indicates an overall improvement in academic performance.
- Arts and Humanities students primarily used GenAI for copywriting whereas Business and Medical students used it for summarizing/ reading.
- Students with paid AI subscriptions spend more time using AI compared to non-paid users.
- AI usage patterns differed across years of study , with Senior students spending the most average hours using GenAI.
- All major categories except STEM consist major portion of students who are beginner in prompt engineering.
- STEM Sophomore are facing high anxiety level.